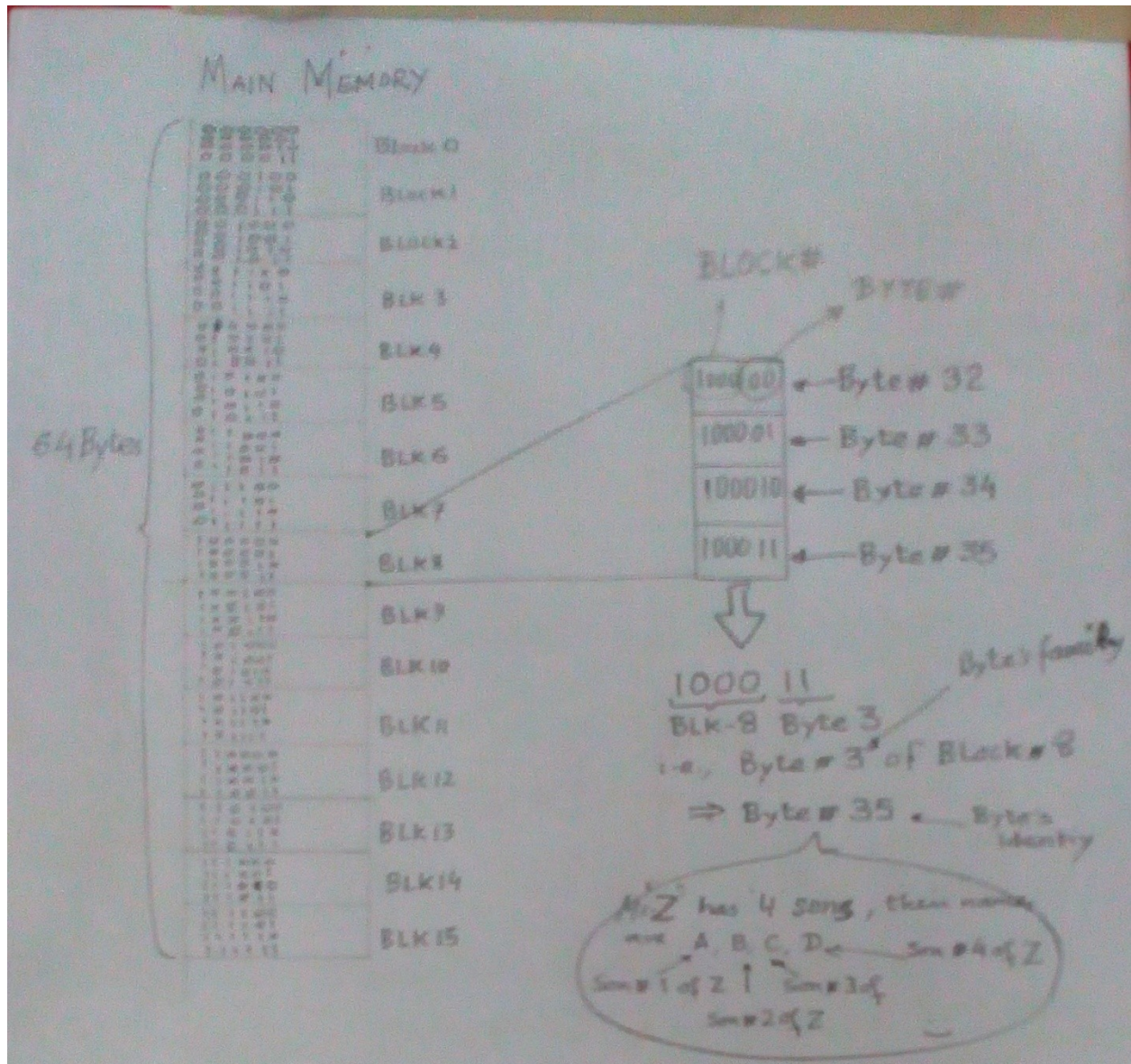
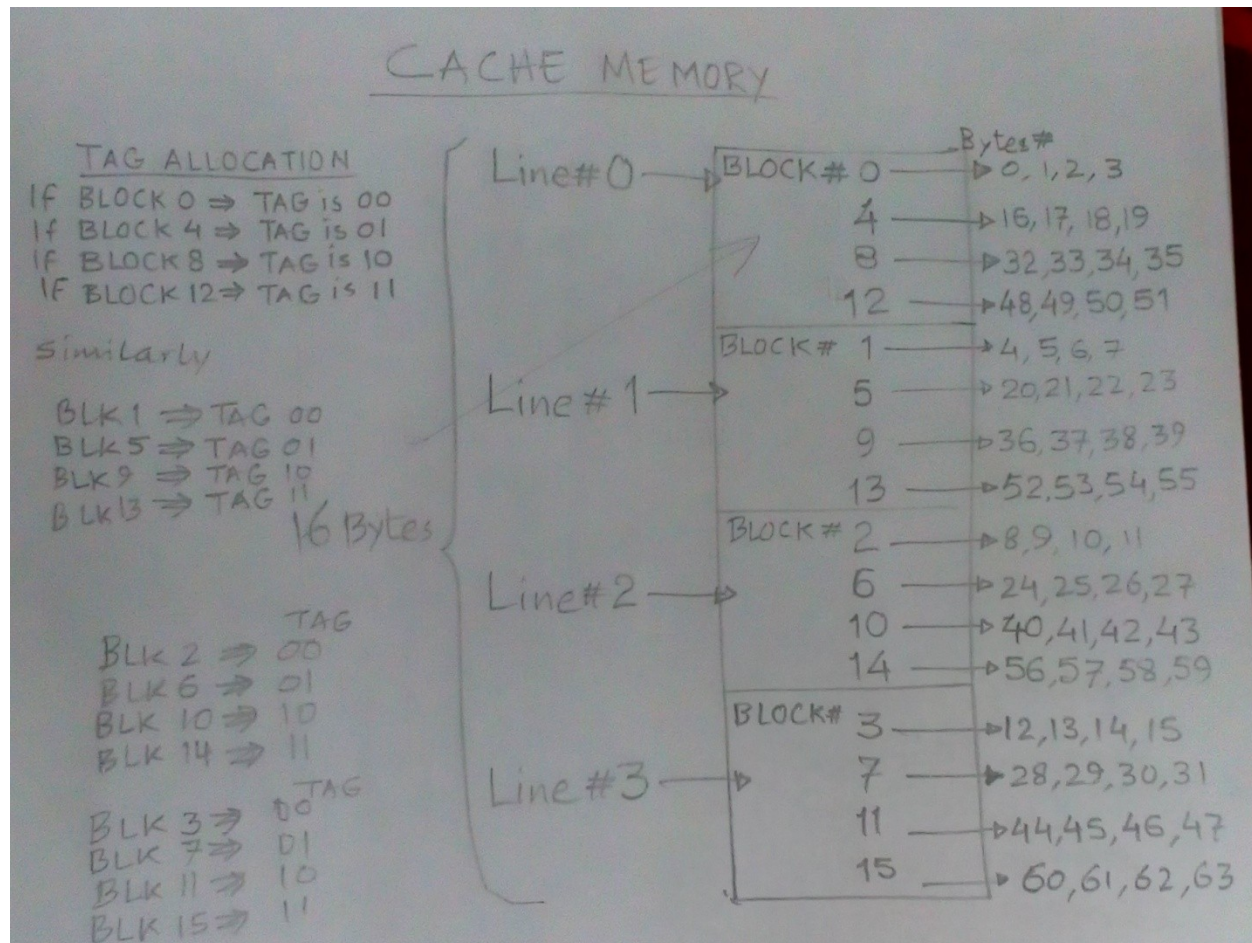


Computer Organization & Assembly Language (CS-215T)

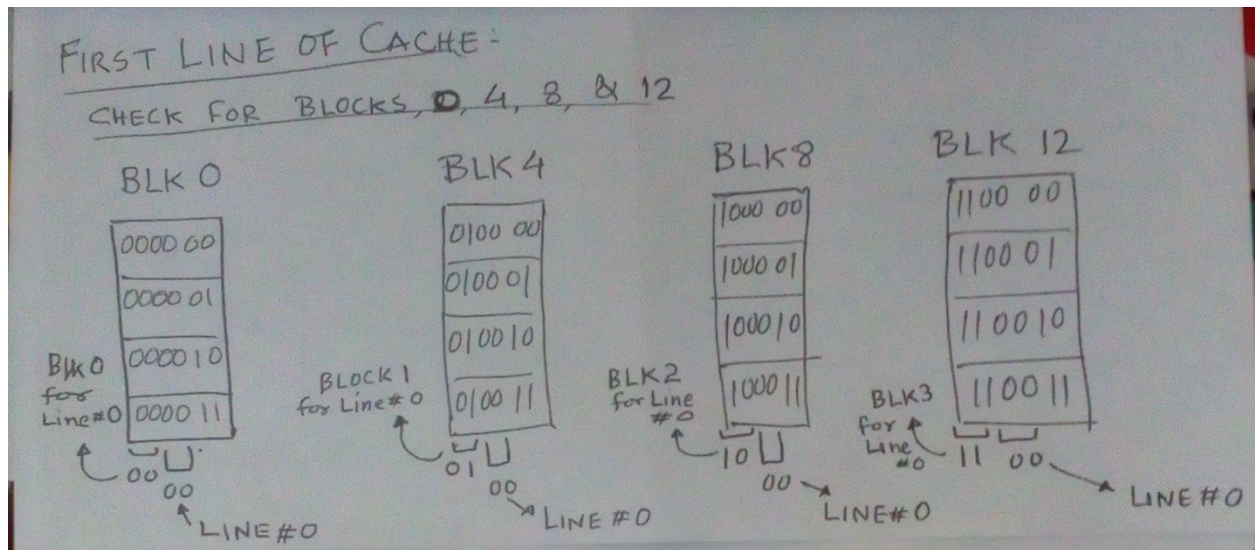
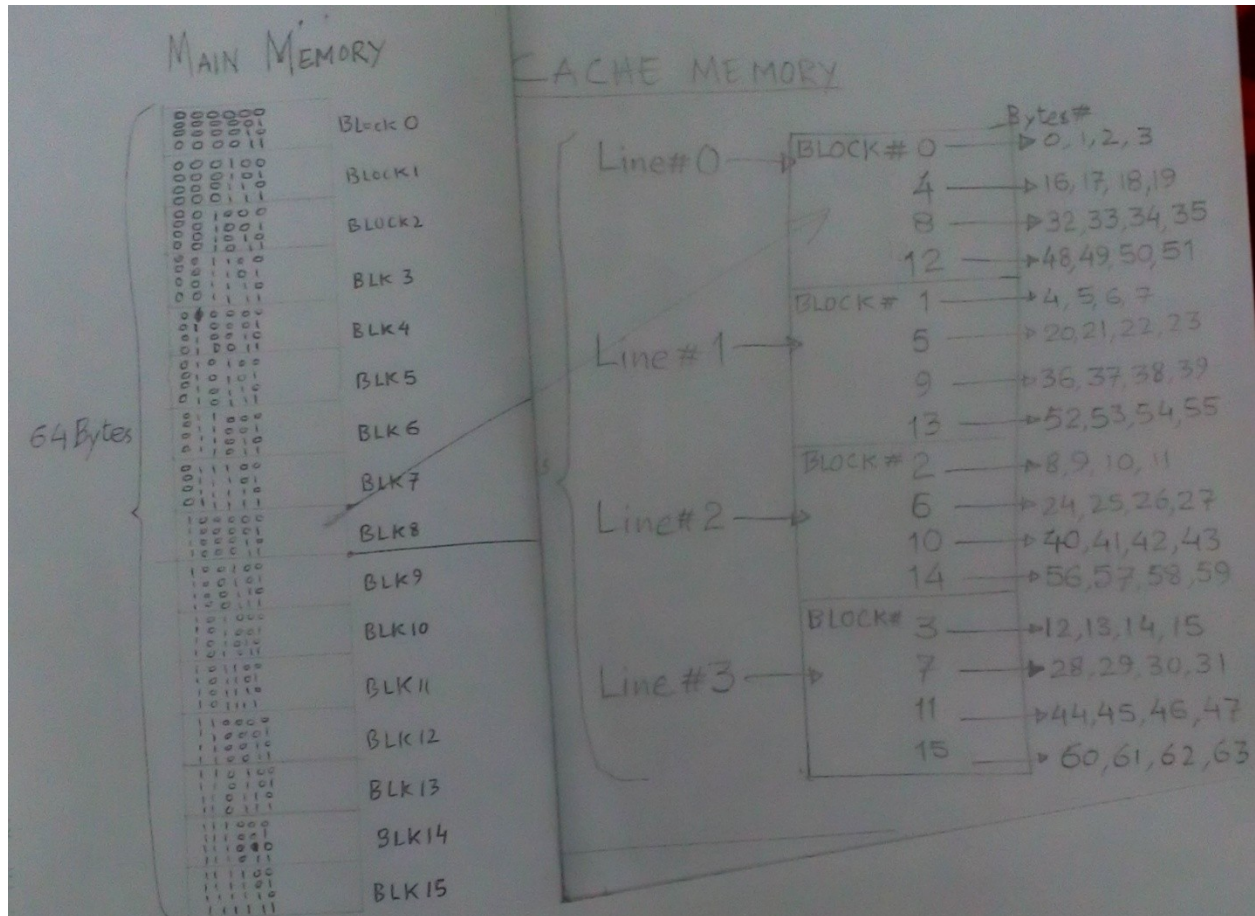
Direct Mapping Technique – Explained by Mr. Shakir Karim



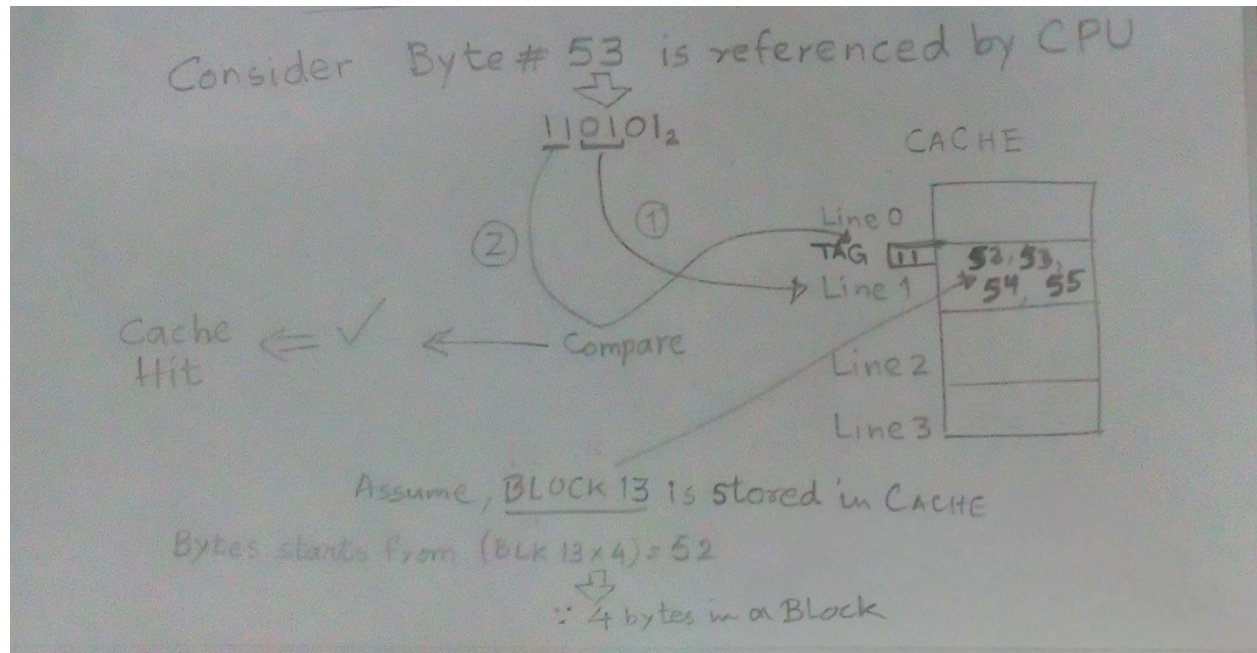
Computer Organization & Assembly Language (CS-215T)
Direct Mapping Technique – Explained by Mr. Shakir Karim



Computer Organization & Assembly Language (CS-215T)
Direct Mapping Technique – Explained by Mr. Shakir Karim



Computer Organization & Assembly Language (CS-215T)
Direct Mapping Technique – Explained by Mr. Shakir Karim



Computer Organization & Assembly Language (CS-215T)

Direct Mapping Technique – Explained by Mr. Shakir Karim

DIRECT MAPPING

ADVANTAGE

Time is saved, since no need to decide where to put the Block in Cache.

Direct mapping is confirmed i.e., Line 0 of Cache can only contain Block 0, 4, 8, or 12. It is fixed that Line 1 can only allow to contains bytes from either Block 1, 5, 9, or 13 and so on.

Consider the previous cache hit example. Suppose, the next byte the CPU wants is Byte # 20. Remember, the Line # 1 of Cache contains Block 13 and TAG bits 11 are associated with Line # 1 of Cache.

Byte # 20
010100
Line # 1
TAG 10
≠ 11
Cache Miss

- Request to Main Memory.
- Block 5 is fetched and a copy is stored in Line # 1 of Cache { Bytes # 20, 21, 22, 23 } with Tag bits 01

Disadvantage of Direct Mapping:

If the next Byte the CPU refer is 54, then Block 5 must be flushed off and Block 13 must be brought back again into Line # 1. It takes time — Extra Time.